$\vec{E} = \frac{\vec{F}}{Q}$ $\quad \vec{D}_0 = \varepsilon_0 \cdot \vec{E}$ $\quad \varepsilon = \varepsilon_0 \cdot \varepsilon_r$ $\quad \varepsilon_0 = 8.854 \cdot 10^{-12}$ As/Vm

$C = \varepsilon_0 \varepsilon_r \frac{S}{l}$ $\quad C = \frac{Q}{U}$ $\quad q$=1.6·10^{-19} C

$Q_1 = Q_2 = Q$ $\quad \frac{1}{C_{uk}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + ... + \frac{1}{C_n}$

$U_1 = U_2 = U$ $\quad C_{uk} = C_1 + C_2 + C_3 + ... + C_n$

$F \cdot l = U \cdot q$ $\quad A = F \cdot l$ $\quad W = U \cdot I \cdot t = U \cdot q$ $\quad \vec{F} = \frac{U \cdot q}{l} = \vec{E} \cdot q$

$E = \frac{Q}{4 \cdot \pi \cdot \varepsilon_0 \cdot \varepsilon_r} \cdot \frac{1}{r^2}$ $\quad F = \left|Q'\right| \cdot E = \frac{1}{4 \cdot \pi \cdot \varepsilon_0 \cdot \varepsilon_r} \cdot \frac{Q \cdot \left|Q'\right|}{r^2}$

$A = \frac{1}{4 \cdot \pi \cdot \varepsilon_0 \cdot \varepsilon_r} \cdot Q \cdot Q' \cdot \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$ $\quad W = \frac{U \cdot Q}{2}$

$\Phi = \frac{I \cdot N}{R_m}$ $\quad R_m = \frac{1}{\mu} \cdot \frac{l}{S}$ $\quad \mu = \mu_r \cdot \mu_0$ $\quad \mu_0 = 4 \cdot \pi \cdot 10^{-7}$ Vs/Am

$B = \frac{\Phi}{S}$ $\quad B = H \cdot \mu$ $\quad H \cdot l = I \cdot N$ $\quad I \cdot N = H_m \cdot l_m + H_z \cdot l_z$

$H = \frac{I}{2 \cdot \pi \cdot r}$ $\quad e = B \cdot l \cdot v$ $\quad F = B \cdot I \cdot l$ $\quad F = B \cdot Q \cdot v$

$e = -N \cdot \frac{d\Phi}{dt}$ $\quad e = -L \cdot \frac{di}{dt}$ $\quad L = \frac{N^2}{R_m}$ $\quad L = \frac{N \cdot \Phi}{I}$

$e_2 = -M_{12} \cdot \frac{di_1}{dt}$ $\quad M_{12} = \frac{N_1 N_2}{R_m}$ $\quad M = k\sqrt{L_1 L_2}$

$W_{mag} = L \frac{I^2}{2}$

$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$ $\quad \varphi = \arctan \frac{X_L - X_C}{R}$

$Y = \sqrt{G^2 + (B_L - B_C)^2} = \sqrt{G^2 + \left(\frac{1}{\omega L} - \omega C \right)^2}$ $\quad \varphi = \arctan \frac{B_L - B_C}{G}$

$P = I_{ef}^2 R = U_{ef}^2 G$ $\quad Q = I_{ef}^2 X = U_{ef}^2 B$ $\quad S = I_{ef}^2 Z = U_{ef}^2 Y$